The Graffiti.pc conjecture $f(G) \geq \lceil (p(G) + b(G) + 1)/2 \rceil$: an equivalent deficiency form, a reduction of the bipartite case to two finite lemmas, and exhaustive verification to $n = 11$

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Abstract

Let G be a connected graph on $n \geq 2$ vertices. Write $f(G)$ for the maximum order of an induced forest, $b(G)$ for the maximum order of an induced bipartite subgraph, and $p(G)$ for the minimum number of vertex-disjoint paths covering $V(G)$. Conjecture 40 of the *Written on the Wall II* (Graffiti.pc) collection asserts $f(G) \geq \lceil (p(G) + b(G) + 1)/2 \rceil$. Introducing the complementary parameters $\tau = n - f$ (decycling number), $o = n - b$ (odd-cycle transversal number), and $\ell = n - p$ (number of edges in a maximum spanning linear forest), we prove the conjecture is equivalent to the linear inequality $\ell + o \geq 2\tau + 1$. For bipartite G this reads $\ell \geq 2\tau + 1$; we reduce this bipartite case, via an incidence-selection flow model, to two explicitly stated lemmas (Lemma A on forced cores and Lemma B on cycle-erasing descent), proving all of the surrounding structure (a generalized Hall criterion, a rooted forest extension bound, a repaired residual-augmentation flow theorem, and a descent argument) and leaving exactly those two lemmas open. On the general side we prove, conditional on the bipartite case, the coefficient-two bound $\ell + 2o \geq 2\tau + 1$, and we isolate a single-vertex reduction property (SV) which, together with the bipartite case, implies the full conjecture. Finally we report exhaustive computational verification: the conjecture holds for all 1,006,700,565 connected graphs on 11 vertices (and all connected graphs on ≤ 10 vertices), with zero violations, under a preregistered dual-engine protocol; SV holds for every connected nonbipartite graph on ≤ 10 vertices.

1 Introduction

For a finite simple graph $G = (V, E)$ on $n = |V|$ vertices, let

$$f(G) = \text{max. order of an induced forest}, \quad b(G) = \text{max. order of an induced bipartite subgraph},$$

and let $p(G)$ be the *path cover number*: the minimum number of vertex-disjoint simple paths (isolated vertices allowed as trivial paths) whose vertex sets partition V. The following appears as Conjecture 40 in DeLaviña's *Written on the Wall II* (WOWII), the conjecture stream of the Graffiti.pc system.

Conjecture 1.1 (WOWII #40)**.** *For every connected graph G on $n \geq 2$ vertices,*

$$f(G) \;\geq\; \left\lceil \frac{p(G) + b(G) + 1}{2} \right\rceil.$$

The specialization $p(G) = 1$ (graphs with a Hamiltonian path) was highlighted early: there the bound reads $f \geq \lceil (b + 2)/2 \rceil = \lfloor b/2 \rfloor + 1$. The conjecture is recorded as open on the maintained

WOWII list (last updated 2026-07-23) and is carried, classified research-open, in the Lean formalization repository `google-deepmind/formal-conjectures` as `GraphConjecture40`. A search of the standard combinatorics literature at the time of writing located no proof, counterexample, or implying theorem (MathSciNet and zbMATH were not searched; this is a search report, not a nonexistence claim). The closest related work is the Hall-type path-cover condition of Lavrov and Vandenbussche [1] and the adjacent super-cyclic literature; see Section 7.

Results. This paper contributes four things.

(i) **An equivalent deficiency form** (Section 2). With $\tau = n - f$, $o = n - b$, $\ell = n - p$, Conjecture 1.1 is equivalent to $\ell(G) + o(G) \geq 2\tau(G) + 1$. We give the exact equality analysis, including a closed margin formula $\mu = n_r - n_{r-1}$ for traceable complete multipartite graphs.

(ii) **A reduction of the bipartite case** (Sections 3–4). For bipartite G the target is $\ell \geq 2\tau + 1$. Through an incidence-selection model we prove a generalized Hall criterion, a rooted forest extension bound, a repaired residual-augmentation flow theorem, and a descent argument, and thereby reduce the bipartite case to exactly two lemmas, both stated precisely and both open.

(iii) **A conditional general bound and a single-vertex reduction** (Section 5). Conditional on the bipartite case we prove the coefficient-two bound $\ell + 2o \geq 2\tau + 1$, and we prove that a single-vertex property (SV) together with the bipartite case implies the full conjecture.

(iv) **Exhaustive verification to** $n = 11$ (Section 6). Under a preregistered, hash-frozen, dual-engine protocol, Conjecture 1.1 holds for every connected graph on ≤ 11 vertices (1,006,700,565 graphs at $n = 11$), with zero violations; SV holds for every connected nonbipartite graph on ≤ 10 vertices.

Two lemmas in the bipartite reduction remain open; the general-to-bipartite step at coefficient one remains open. The paper states exactly what is proved, what is conditional, and what is verified only computationally.

2 The deficiency form

Throughout, G is connected with $n \geq 2$ vertices. Define the three *deficiency parameters*

$$\tau(G) = n - f(G), \qquad o(G) = n - b(G), \qquad \ell(G) = n - p(G).$$

Here τ is the *decycling number* (minimum size of a feedback vertex set: deleting a maximum induced forest leaves a minimum FVS), o is the *odd-cycle transversal number* (deleting a maximum induced bipartite subgraph leaves a minimum odd-cycle transversal), and ℓ is the maximum number of edges in a spanning *linear forest* (a disjoint union of paths): a path cover by p paths has exactly $n - p$ edges, so maximizing edges is minimizing paths.

Theorem 2.1 (Equivalence)**.** *For every connected graph G on $n \geq 2$ vertices,*

$$f(G) \geq \left\lceil \frac{p(G) + b(G) + 1}{2} \right\rceil \quad \Longleftrightarrow \quad \ell(G) + o(G) \;\geq\; 2\tau(G) + 1.$$

In particular, for bipartite G (where $o = 0$) the conjecture is $\ell(G) \geq 2\tau(G) + 1$.

Proof. The ceiling inequality $f \geq \lceil (p+b+1)/2 \rceil$ is equivalent, over the integers, to $2f \geq p+b+1$: indeed $f \geq \lceil m/2 \rceil$ iff $2f \geq m$ for any integer m, since $2f$ is even and $\lceil m/2 \rceil$ is the least integer t with $2t \geq m$. Substitute $f = n-\tau$, $b = n-o$, $p = n-\ell$:

$$2(n-\tau) \;\geq\; (n-\ell)+(n-o)+1 \iff -2\tau \geq -\ell - o + 1 \iff \ell + o \;\geq\; 2\tau + 1. \qquad \square$$

We verified Theorem 2.1 numerically on all 995 connected graphs with $2 \leq n \leq 7$ (checker `phase0.c`, Section 6): the two sides agreed in every case.

Margin and tightness. Define the *linear margin*

$$\mu(G) \;:=\; \ell(G) + o(G) - \big(2\tau(G) + 1\big).$$

Conjecture 1.1 is the assertion $\mu(G) \geq 0$. We call G *linear-tight* when $\mu(G) = 0$. One must not conflate linear tightness with equality in the original ceiling statement: because of the parity of the ceiling, equality $f = \lceil (p+b+1)/2 \rceil$ holds precisely when $\mu(G) \in \{0, 1\}$. Thus "ceiling-equality" is the union of the linear-tight shell $\mu = 0$ and the neighboring shell $\mu = 1$.

Proposition 2.2 (Equality families)**.** *Let $G = K_{n_1,\ldots,n_r}$ be complete multipartite with $r \geq 2$ parts ordered $n_1 \leq \cdots \leq n_r$ and $n = \sum_i n_i$. Then*

$$p(G) = \max(1,\, 2n_r - n), \qquad \mu(G) = \begin{cases} n_r - n_{r-1}, & p = 1 \quad (\textit{i.e. } 2n_r \leq n+1), \\ (n - n_r) - n_{r-1} + 1, & p > 1 \quad (\textit{i.e. } 2n_r > n+1). \end{cases}$$

Consequently G is linear-tight ($\mu = 0$) iff it is traceable and its two largest parts are equal ($n_{r-1} = n_r$); this includes K_n and $K_{a,a}$. And $\mu = 1$ for every $K_{a,b}$ with $a < b$ and for every star $K_{1,m}$ with $m \geq 2$. Separately, C_3, C_4, and K_2 are linear-tight.

Proof. An induced subgraph of $K_{n_1,\ldots,n_r}$ is again complete multipartite, hence a forest iff it is edgeless or a star; the largest induced forest is one vertex plus a largest part, so $f = n_r + 1$ and $\tau = n - n_r - 1$. An induced subgraph is bipartite iff at most two parts are nonempty, so $b = n_r + n_{r-1}$ and $o = n - n_r - n_{r-1}$. A complete multipartite graph is traceable iff the largest part does not exceed the rest by more than one, i.e. $2n_r \leq n+1$; otherwise the $2n_r - n$ excess vertices of the largest part each start a new path, so $p = \max(1, 2n_r - n)$.

If $p = 1$ then $\ell = n - 1$ and

$$\mu = (n-1) + (n - n_r - n_{r-1}) - \big(2(n - n_r - 1) + 1\big) = n_r - n_{r-1}.$$

If $p > 1$ then $\ell = n - p = n - (2n_r - n) = 2(n - n_r)$ and

$$\mu = 2(n-n_r) + (n - n_r - n_{r-1}) - \big(2(n - n_r - 1) + 1\big) = (n - n_r) - n_{r-1} + 1.$$

The two formulas agree on the boundary $2n_r = n+1$. For $K_{a,b}$ with $a < b$: if $b = a+1$ (traceable) the first formula gives $\mu = b - a = 1$; if $b \geq a+2$ the second gives $\mu = (a+b-b) - a + 1 = 1$; so every $K_{a,b}$ with $a < b$ has $\mu = 1$. For a star $K_{1,m}$ ($m \geq 2$), $n_r = m, n_{r-1} = 1, n = m+1$, and the $p > 1$ formula gives $\mu = (m+1-m) - 1 + 1 = 1$ (matching $p = m-1$). For linear tightness: in the $p > 1$ regime $(n - n_r) - n_{r-1} + 1 \geq 1$ because $n_{r-1} \leq n - n_r$ (the second-largest part is at most the sum of all non-largest parts), so $\mu = 0$ forces $p = 1$ and then $n_{r-1} = n_r$; conversely $p = 1$ with $n_{r-1} = n_r$ gives $\mu = 0$. Finally direct evaluation gives $\mu = 0$ for K_2 ($\ell = 1, o = 0, \tau = 0$), C_3 $(2, 1, 1)$, and C_4 $(3, 0, 1)$. $\square$

These formulas locate the extremal graphs of the deficiency form: the traceable complete multipartite graphs with equal top two parts are exactly linear-tight, and the whole complete bipartite family $K_{a,b}$ $(a \neq b)$ together with the stars sits on the neighboring $\mu = 1$ shell. They cross-check against the gold invariants of Section 6: $K_{10}\,(f, b, p) = (2, 2, 1)$, $K_{5,5} = (6, 10, 1)$, $K_{1,9} = (10, 10, 8)$, $C_{10} = (9, 10, 1)$.

3 The bipartite case: reduction to two lemmas

Fix a connected nontrivial bipartite graph G with bipartition $V = L \cup R$. The target is $\ell(G) \geq 2\tau(G) + 1$. If $\tau(G) = 0$ then G is a nontrivial forest, has an edge, and that edge (with all other vertices as trivial paths) is a spanning linear forest of $\geq 1 = 2\tau + 1$ edges; so assume $\tau \geq 1$.

3.1 The incidence-selection model

Let S be a minimum feedback vertex set, $|S| = \tau$. Then $H := G - S$ is a forest (deleting an FVS destroys all cycles). For a vertex set X we write $G[H \cup X]$ for the subgraph induced on $H \cup X$. We build a capacitated selection problem on the S–H incidences.

Definition 3.1 (Selection model)**.** Fix an edge $e = (x, y)$ of H (its endpoints lie in opposite bipartition classes). Each $s \in S$ has *demand* 2. Each $h \in H$ has *capacity* $c(h) = 2$, except $c(x) = c(y) = 1$. A *feasible selection* J assigns to each $s \in S$ two distinct S–H incidences at s so that each $h \in H$ receives at most $c(h)$ selected incidences. A *selected cycle* of J is a cyclic component of the incidence graph J.

The point of the model is the following counting fact, proved in Section 3.5: an *acyclic* feasible selection J (with $J \cup \{e\}$ acyclic) yields a spanning linear forest of G with exactly $2|S| + 1$ edges, hence $\ell \geq 2\tau + 1$. So the bipartite target reduces to the existence of an acyclic feasible selection for some fixed edge e.

We record the canonical semantics of a cycle-erasing move once, and use it throughout.

Definition 3.2 (CES move)**.** A toggle from a feasible selection J to a feasible selection J' is *cycle-erasing (CES)* for a designated selected cycle C of J when (i) C is not a selected cycle of J', and (ii) every selected cycle of J' was already a selected cycle of J. This is a semantic condition (compare the selected-cycle sets before and after); it implies a strict decrease in the number of selected cycles. CES is *not* defined merely as "fewer selected cycles"; a strict decrease alone would already suffice for termination of the descent (Section 3.5), but the stronger no-new-cycle form is what the mechanism produces.

3.2 Hereditary tightness and the Hall criterion

Lemma 3.3 (Hereditary FVS tightness)**.** *For every $X \subseteq S$, $\tau(G[H \cup X]) = |X|$.*

Proof. X is an FVS of $G[H \cup X]$ (deleting it leaves H, a forest), so $\tau(G[H \cup X]) \leq |X|$. Conversely, suppose D is an FVS of $G[H \cup X]$ with $|D| < |X|$. Then $(S \setminus X) \cup D$ is an FVS of G: deleting it leaves a subgraph of $G[H \cup X] - D$, which is a forest. Its size is $|S| - |X| + |D| < |S|$, contradicting minimality of S. Hence $\tau(G[H \cup X]) = |X|$. □

Encode feasibility as a flow. Build a network N with a source σ; arcs $\sigma \to s$ of capacity 2 for each $s \in S$; an arc $s \to h$ of capacity 1 for each incidence; and arcs $h \to$ sink of capacity $c(h)$. An

integral flow saturating every $\sigma \to s$ is exactly a feasible selection. For $X \subseteq S$ put

$$\rho(X) = \sum_{h \in H} \min\big(c(h), \deg_X(h)\big), \qquad \delta(X) = \rho(X) - 2|X|,$$

where $\deg_X(h)$ is the number of neighbours of h in X.

Lemma 3.4 (Generalized Hall criterion)**.** *A feasible selection exists iff $\rho(X) \geq 2|X|$ for every $X \subseteq S$.*

Proof. Consider any σ–sink cut (A, B) with $\sigma \in A$, sink $\in B$, and set $X = S \cap A$. The forward capacity is: source arcs $\sigma \to s$ for $s \in S \cap B$ cost 2 each, total $2(|S| - |X|)$; an incidence arc $s \to h$ with $s \in A, h \in B$ can be forward only for $s \in X$, cost 1; a sink arc $h \to$ sink with $h \in A$ costs $c(h)$. For each h, its contribution is minimized by placing it on the cheaper side, giving $\min(c(h), \deg_X(h))$. Hence the cheapest cut with a given X has capacity $2(|S| - |X|) + \rho(X) = 2|S| + \delta(X)$, and the global minimum cut is $2|S| + \min_X \delta(X)$. By max-flow/min-cut the maximum flow equals $\min(2|S|, 2|S| + \min_X \delta(X))$, which is $2|S|$ (a feasible selection) iff $\min_X \delta(X) \geq 0$. □

An incidence $a = (s, h)$ is *forced* if it belongs to every feasible selection; the *forced core* is the graph of all forced incidences.

Lemma 3.5 (Forced $\iff$ tight-set exposed, F1)**.** *Fix a feasible selection. An incidence $a = (s, h)$ is forced iff some ρ-tight set X (i.e. $\delta(X) = 0$) exposes it: $s \in X$ and $\deg_X(h) \leq c(h)$.*

Proof. Deleting the arc a from N lowers $\rho(X)$ by exactly one iff $s \in X$ and $\deg_X(h) \leq c(h)$. If a is forced then $N - a$ has no feasible flow, so by Lemma 3.4 some X has $\rho_{N-a}(X) < 2|X|$; since $\rho(X) \geq 2|X|$ held, X was tight and a lowered it, so X exposes a. Conversely, if a tight X exposes a, then in $N - a$ we have $\rho_{N-a}(X) = 2|X| - 1 < 2|X|$, so no feasible selection omits a. □

Lemma 3.6 (Tight-set lattice)**.** *δ is submodular and nonnegative on subsets of S; hence if X, Y are tight so are $X \cap Y$ and $X \cup Y$.*

Proof. Each map $X \mapsto \min(c(h), \deg_X(h))$ is submodular, being the composition of the concave nondecreasing function $t \mapsto \min(c(h), t)$ with the modular map $X \mapsto \deg_X(h)$; hence ρ and $\delta(X) = \rho(X) - 2|X|$ are submodular, and feasibility gives $\delta \geq 0$. If $\delta(X) = \delta(Y) = 0$ then $0 \leq \delta(X \cap Y) + \delta(X \cup Y) \leq \delta(X) + \delta(Y) = 0$, forcing both to vanish. □

3.3 Feasibility for every fixed edge

We show a feasible selection actually exists—for *every* choice of the distinguished edge e. This uses a rooted forest bound of independent interest.

Lemma 3.7 (Rooted forest extension)**.** *Let G be bipartite, H an induced forest, and X an independent set contained in one bipartition class. Fix an edge $e = ab$ of H; give a, b capacity 1 and every other H-vertex capacity 2, and put $\rho_e(X) = \sum_h \min(c_e(h), \deg_X(h))$. Then*

$$\tau(G[H \cup X]) \;\leq\; \lfloor \rho_e(X)/2 \rfloor .$$

Proof. Let r be the endpoint of e in the bipartition class opposite X; the other endpoint lies in X's class and so has no X-neighbour. Thus r is the only e-endpoint that can have X-neighbours. Induct on $k = \#\{h \in H \setminus \{r\} : \deg_X(h) \geq 2\}$.

Step ($k > 0$). Pick $h \in H \setminus \{r\}$ with $\deg_X(h) \geq 2$; then h is neither endpoint of e, so $c_e(h) = 2$. Put $K = G[H \cup X]$, $K' = K - h$. Then $H - h$ is a forest containing e, so the lemma applies to K', and deleting h lowers ρ_e by $\min(2, \deg_X(h)) = 2$. If D is an FVS of K' then $D \cup \{h\}$ is an FVS of K, so $\tau(K) \leq 1 + \lfloor (\rho_e^K(X) - 2)/2 \rfloor = \lfloor \rho_e^K(X)/2 \rfloor$.

Base ($k = 0$). Every $h \in H \setminus \{r\}$ has ≤ 1 X-neighbour. *Case 1* ($\deg_X(r) \leq 1$): every h has ≤ 1 X-neighbour, so $\rho_e(X)$ equals the number of X–H incidences. Let $D = \{x \in X : \deg_H(x) \geq 2\}$; each contributes ≥ 2 incidences, so $2|D| \leq \rho_e(X)$. In $K - D$ every surviving X-vertex has degree ≤ 1; adjoining leaves to the forest H keeps it a forest, so $K - D$ is a forest and $\tau(K) \leq |D| \leq \lfloor \rho_e(X)/2 \rfloor$. *Case 2* ($\deg_X(r) \geq 2$): let m be the number of X-to-$(H \setminus \{r\})$ incidences; each non-root H-vertex contributes ≤ 1 and r contributes $c_e(r) = 1$, so $\rho_e(X) = m + 1$. Let $Q = \{x \in X : \deg_{H-r}(x) \geq 2\}$, $q = |Q|$, so $2q \leq m$. If $m \geq 2q + 1$, take $D = Q \cup \{r\}$: after deleting r, each $x \notin Q$ has ≤ 1 remaining H-neighbour, so $K - D$ is a forest and $|D| = q + 1 \leq \lfloor (m + 1)/2 \rfloor = \lfloor \rho_e(X)/2 \rfloor$. If $m = 2q$, take $D = Q$: every $x \in Q$ has exactly two non-root incidences and every $x \notin Q$ none, so surviving X-vertices have degree ≤ 1 (only possibly to r) and $K - D$ is a forest, with $|D| = q = \lfloor (2q + 1)/2 \rfloor = \lfloor \rho_e(X)/2 \rfloor$. □

Corollary 3.8 (Initial feasibility for every e)**.** *For every $X \subseteq S$ and every fixed edge $e \in E(H)$, $\rho_e(X) \geq 2|X|$; hence a feasible selection exists for every distinguished edge e.*

Proof. Split X into $X_L = X \cap L$ and $X_R = X \cap R$, each independent. Every $h \in H \cap L$ has neighbours only in X_R and every $h \in H \cap R$ only in X_L, so the supports are disjoint and $\rho_e(X) = \rho_e(X_L) + \rho_e(X_R)$. If $\rho_e(X) < 2|X|$ then one side, say X_L, has $\rho_e(X_L) < 2|X_L|$; then Lemma 3.7 gives $\tau(G[H \cup X_L]) \leq \lfloor \rho_e(X_L)/2 \rfloor < |X_L|$, contradicting hereditary tightness (Lemma 3.3). Thus $\rho_e(X) \geq 2|X|$ for all $X \subseteq S$, and Lemma 3.4 gives feasibility. □

3.4 Parity and residual augmentation

Lemma 3.9 (Fixed-edge parity)**.** *The fixed edge $e = (x, y)$ lies on no selected cycle, and $J \cup \{e\}$ is acyclic iff J is.*

Proof. If $J \cup \{e\}$ had a cycle through e, deleting e leaves an x–y path of S–H incidence edges, which alternates $H, S, H, \ldots$ and so has even length. But x, y are adjacent in bipartite G, hence in opposite classes, so every x–y path has odd length — a contradiction. □

The next theorem is the corrected engine of the descent. (An earlier symmetric-difference sketch was superseded; the argument below is a direct max-flow augmenting-path proof.)

Theorem 3.10 (Residual augmentation)**.** *Let J be a feasible selection and $a = (s, h)$ a selected incidence. Delete the arc a from N entirely; call the result $N - a$. Then a feasible selection omitting a exists iff $N - a$ admits a flow of value $2|S|$ iff there is a simple residual σ–sink augmenting path in $N - a$. Consequently a is avoidable iff it is avoidable by a single residual augmentation.*

Proof. J is a value-$2|S|$ integral flow in N. Form $J - a$ by deleting one complete unit-flow path $\sigma \to s \to h \to$ sink (the unit routed through a): the flow on $\sigma \to s$ drops from 2 to 1; the arc a is absent from $N - a$; the flow on $h \to$ sink drops by 1; all other arc flows are unchanged. Conservation is preserved at s (inflow and outflow both drop by 1) and at h (likewise); every other vertex is untouched. So $J - a$ is a valid integral flow of value $2|S| - 1$ in $N - a$, in which $\sigma \to s$ is the only unsaturated source arc.

A feasible selection omitting a is a value-$2|S|$ flow in $N - a$ (it saturates every $\sigma \to s$ and never uses a). Conversely, by the augmenting-path theorem applied to $J - a$, the network $N - a$ admits

a larger (value-$2|S|$) flow iff there is a simple residual σ–sink path from the deficient s; augmenting along it saturates $\sigma \to s$ and yields a feasible selection. Its incidence portion (delete the source/sink arcs) is exactly an alternating augmenting path from s to an H-vertex of residual capacity. The released arc a is absent from $N - a$, so no residual path can re-add it. □

Corollary 3.11. *There is no incidence that is globally avoidable yet admits no single residual augmentation avoiding it; "avoidable" and "one-move avoidable" coincide.*

3.5 Descent, and the reduction to two lemmas

Lemma 3.12 (Descent)**.** *If, from every feasible selection with a selected cycle, some CES toggle exists, then repeatedly applying CES toggles terminates in an acyclic feasible selection.*

Proof. A CES toggle destroys the target cycle and creates none, so it strictly lowers the (nonnegative integer) number of selected cycles. A strictly decreasing sequence of nonnegative integers terminates; the terminal selection is acyclic. □

We now name the two properties that a CES toggle needs, and package the whole bipartite case. Write $F2$ for the statement that an acyclic feasible selection exists for some fixed edge e.

Theorem 3.13 (Reduction of the bipartite case)**.** *Suppose the following two lemmas hold for all connected nontrivial bipartite graphs.*

- ***A.*** *(Forced core acyclic.) In the model of Definition 3.1, the forced core contains no cycle.*
- ***B.*** *(Cycle-erasing descent.) Every selected cycle that contains an avoidable (non-forced) incidence admits a CES toggle.*

Then every connected nontrivial bipartite graph satisfies $\ell \geq 2\tau + 1$, i.e. Conjecture 1.1 holds in the bipartite case.

Proof. Take $\tau \geq 1$ (the $\tau = 0$ case was handled at the start of Section 3). We first check $E(H) \neq \varnothing$, so a fixed edge exists. If $E(H) = \varnothing$, take $X = \{s\}$ for any $s \in S$: then $G[H \cup \{s\}]$ has edges only at s (a star forest), so $\tau(G[H \cup \{s\}]) = 0$, contradicting hereditary tightness (Lemma 3.3), which gives 1. So fix $e \in E(H)$.

By Corollary 3.8 a feasible selection exists. By Lemma A the forced core is acyclic, so every selected cycle contains an avoidable incidence: forced incidences belong to every feasible selection, so a selected cycle all of whose incidences were forced would be a cycle of the forced core, contradicting acyclicity. Hence Lemma B applies to every selected cycle and supplies a CES toggle, and the descent (Lemma 3.12) terminates in a feasible selection J with no selected cycle. By Lemma 3.9, $J \cup \{e\}$ is acyclic. Now $J \cup \{e\}$ has maximum degree 2: every $s \in S$ has selected degree 2; every $h \in H$ has degree $\leq c(h) \leq 2$; and x, y keep degree ≤ 2 because $c(x) = c(y) = 1$ leaves one unit for the edge e. An acyclic graph of maximum degree 2 is a disjoint union of paths, i.e. a linear forest. It has $2|S|$ incidence edges plus the edge e, and it spans $S \cup H = V$; every s has degree 2, every internal vertex degree ≤ 2. Hence it is a spanning linear forest with $2|S| + 1$ edges, so $\ell \geq 2|S| + 1 = 2\tau + 1$. □

Remark 3.14. Theorem 3.13 isolates the open content precisely. All of Sections 3.2–3.5 — the Hall criterion, the F1 characterization, the tight-set lattice, the rooted forest bound and feasibility for every e, fixed-edge parity, the residual-augmentation flow theorem, and the descent — are proved unconditionally. What remains for the bipartite case is exactly Lemma A and Lemma B.

4 The two open lemmas

We state the two lemmas in the strongest proved-adjacent forms and record the footholds that constrain any proof.

4.1 Lemma A: the forced core is a forest

It is essential that Lemma A use hereditary tightness, not merely Hall feasibility. The abstract even-cycle system (S_i adjacent to H_i, H_{i+1}, demand 2, capacity 2) is Hall-feasible yet its unique feasible selection is the whole cycle, so its forced core is a cycle; there $\tau(G[H \cup S]) = 1 < |S|$, i.e. hereditary tightness fails. Thus Lemma A is exactly the implication

$$\text{hereditary FVS tightness for all } X \subseteq S \implies \text{forced core acyclic},$$

and this implication is open in general. We state the lemma without a connectedness hypothesis, since none of the footholds uses it and the disconnected form implies the connected one.

Open Problem 4.1 (Lemma A°)**.** Let G be a finite simple bipartite graph, *not necessarily connected*, S a minimum FVS, $H = G - S$, and e a fixed edge of H with capacities as in Definition 3.1. Prove that the forced incidence core is acyclic.

A connected instance is a special case, so Lemma A° implies the connected Lemma A used in Theorem 3.13.

The following foothold reduces the problem to cycles with an external neighbour.

Lemma 4.2 (Closed-neighbourhood subcase, A-I)**.** *Suppose the forced core has a cycle F with S-side P and H-side Q ($|P| = |Q| = m$). If $N_H(P) \subseteq Q$, then hereditary tightness is contradicted; hence no such F exists.*

Proof. In $G[H \cup P]$ pick any $q_0 \in Q$ and set $D = Q \setminus \{q_0\}$, $|D| = m - 1$. Any cycle of $G[H \cup P] - D$ lies in $H - D$ (impossible, an induced subgraph of the forest H) or passes through some $s \in P$. The cycle F alternates S- and H-vertices, while the bipartition classes alternate along every edge; hence all S-positions of the even cycle F lie in one class, so P is contained in one bipartition class. Therefore s has no neighbours in P, and P–$(S \setminus P)$ edges are absent from $G[H \cup P]$; with $N_H(s) \subseteq Q$, deleting $Q \setminus \{q_0\}$ leaves each $s \in P$ of degree ≤ 1, hence on no cycle. So D is an FVS of size $m - 1$, giving $\tau(G[H \cup P]) \leq m - 1 < m$, contradicting Lemma 3.3. □

Two saturation facts follow from tightness; both are used to pin down the canonical carriers.

Lemma 4.3 (Tight-set saturation)**.** *For a tight X ($\rho(X) = 2|X|$), every feasible J and every $h \in H$ have $\mathrm{load}_X^J(h) = \min(c(h), \deg_X(h))$, where $\mathrm{load}_X^J(h)$ counts selected X–h incidences.*

Proof. Termwise $\mathrm{load}_X^J(h) \leq \min(c(h), \deg_X(h))$, while

$$\sum_h \mathrm{load}_X^J(h) = 2|X| = \rho(X) = \sum_h \min\big(c(h), \deg_X(h)\big);$$

equal sums of termwise-dominated quantities force equality in every term. □

Lemma 4.4 (Common tight carrier)**.** *For s on a forced cycle, the two forced incidences at s share a single inclusion-minimal tight exposing set A_s, and A_s is the unique minimum tight set containing s.*

Proof. Let X^-, X^+ be inclusion-minimal tight sets exposing the two forced incidences f^-, f^+ at s; these exist by finiteness (Lemma 3.5 gives some tight exposer, and a smallest one exists). By Lemma 3.6, $X^- \cap X^+$ is tight; it contains s and, having coordinatewise degree $\leq$ that of each exposer, still exposes both, so minimality gives $X^- \cap X^+ = X^- = X^+ =: A_s$. For any tight $Y \ni s$, $A_s \cap Y$ is tight, contains s, and exposes a forced incidence at s, so $A_s \cap Y = A_s$, i.e. $A_s \subseteq Y$. □

Lemma 4.5 (Punctured tightness)**.** *Let $B_s = A_s \setminus \{s\}$. Then B_s is tight; the only H-neighbours v of s with $\deg_{A_s}(v) \leq c(v)$ are the two forced ends q^-, q^+; and every other H-neighbour h of s has $\deg_{A_s}(h) \geq c(h) + 1$.*

Proof. Removing s lowers $\min(c(v), \deg_{A_s}(v))$ by one exactly at the neighbours v with $\deg_{A_s}(v) \leq c(v)$; let k be their number. The forced ends q^-, q^+ are two such vertices, so $k \geq 2$; and $\rho(B_s) = \rho(A_s) - k = 2|A_s| - k$, so $\delta(B_s) = 2 - k \geq 0$ gives $k \leq 2$. Hence $k = 2$, $\delta(B_s) = 0$, the two are exactly q^-, q^+, and every other H-neighbour is oversubscribed. □

Lemma 4.6 (Single-blocker subcase)**.** *Let h be an external H-neighbour of s with $c(h) = 2$ and $U_h = N_G(h) \cap B_s$. If every $u \in U_h$ has at most one H-neighbour other than h, then $\tau(G[H \cup U_h]) \leq 1 < |U_h|$, a contradiction.*

Proof. By Lemma 4.5, $\deg_{A_s}(h) \geq c(h) + 1 = 3$; since $s \in N(h)$, the remaining neighbours give $|U_h| = |N(h) \cap B_s| \geq 2$. All of U_h share the neighbour h, so lie in one bipartition class and induce no edges among themselves. In $G[H \cup U_h] - h$ each $u \in U_h$ then has degree ≤ 1 and lies on no cycle, and $H - h$ is a forest; so $\{h\}$ is an FVS, giving $\tau(G[H \cup U_h]) \leq 1 < 2 \leq |U_h|$, contradicting hereditary tightness. □

Finally, a normalization that lets one assume S equals the union of carriers.

Lemma 4.7 (T_F normalization)**.** *Suppose a Lemma-A° instance has a forced-core cycle F with S-side P. Put $T_F = \bigcup_{s \in P} A_s$ and restrict to $G' = G[H \cup T_F]$, $S' = T_F$ (same H, edge e, capacities). Then $G' - S' = H$; S' is a minimum FVS of G'; hereditary tightness holds for every $X \subseteq T_F$; the restricted system is feasible; every A_s stays tight and exposes both forced edges at s; and every edge of F remains forced in G'. Hence a minimal counterexample to Lemma A° may be assumed to satisfy $S = T_F$.*

Proof. (i) For every $X \subseteq T_F$, the induced subgraphs $G'[H \cup X]$ and $G[H \cup X]$ are literally the same graph (both are induced on $H \cup X$ in G), so $\tau(G'[H \cup X]) = \tau(G[H \cup X]) = |X|$ verbatim from Lemma 3.3 applied to G; hereditary tightness holds on all $X \subseteq T_F$. (ii) Taking $X = T_F$ gives $\tau(G') = |T_F|$; and T_F is an FVS of G' (deleting it leaves H, a forest), hence a minimum FVS. (iii) The quantity ρ is intrinsic to $(X, H,$ capacities, X-to-H edges), all unchanged in G'; so the Hall inequality $\rho(X) \geq 2|X|$ for all $X \subseteq S$ restricts to all $X \subseteq T_F$, and Lemma 3.4 gives feasibility of the restricted system. (iv) Forcedness is *not* monotone under restriction: removing the demand of S-vertices outside T_F can create new feasible selections. Instead, each F-edge is re-forced via Lemma 3.5: its carrier $A_s \subseteq T_F$ is still tight (same ρ value) and still exposes it (same degrees and capacities), so no feasible selection of G' omits it, and F remains in the forced core of G'. Connectedness may be lost under restriction — which is exactly why Lemma A° is stated without it. □

Warning witnesses. Three certified instances (found and independently recomputed) bracket the false localizations and show why the remaining case is genuinely open. Each eliminates a tempting shortcut. None of them is a counterexample to Lemma A°: each is an instance in which

hereditary tightness *fails* at a specific location (so its forced cycle is permitted), and each is used only to refute a proposed rule for *where* that violation must sit — not to exhibit a hereditarily tight instance with a forced cycle, which is exactly what Lemma A° forbids.

- **O-56** (capacity 2, acyclic quotient). Here the hereditary violation sits inside B_s and a *single* external H-vertex blocks it: this is exactly the single-blocker subcase of Lemma 4.6 ($A_s = \{0,2,3\}$, $B_s = \{2,3\}$ tight, forced ends at $\{4,5\}$, external neighbour 6 with $\deg_{A_s}(6) = 3$; the incidences $2 \sim \{6,7\}$, $3 \sim \{6,8\}$). It shows the subcase is nonempty but does not by itself close the general case.

- **O-63** (capacity 2, cyclic quotient). Here every vertex of U_h has incidence-degree 3, the violation is the *entire* $B_s = \{u,a,b\}$, and the blocker $\{h,w\}$ has size 2. This refutes both "every B_s-vertex has incidence-degree 2" and "incidence-degree ≥ 3 forces a private cycle": the blocking set can exceed a single external vertex, and high degree alone does not force a cycle in the forest quotient.

- **O-64** (capacity 1). Explicitly: $S = \{s,t,u\}$; H is the tree $q_1 - r - q_2$ together with $q_1 - y$, $y - h$, $y - w$; fixed edge $e = (h,y)$ with $c(h) = c(y) = 1$; adjacencies $s \sim q_1, q_2, h$, $t \sim q_1, q_2$, $u \sim h, w$. The unique feasible selection sends $s,t \to \{q_1,q_2\}$ and $u \to \{h,w\}$, so u saturates the capacity-1 vertex h, s is pushed off h, and $s - q_1 - t - q_2 - s$ is a forced cycle. Here the carrier puncture $B_s = \{u\}$ is Hall-tight yet $\tau(G[H \cup B_s]) = 1 = |B_s|$: it carries *no* hereditary violation. The first violation sits on the cycle's own S-side $P = \{s,t\}$ ($\tau(G[H \cup P]) = 1 < 2$). This refutes B_s-localization and shows that capacity-1 and capacity-2 external neighbours are separate branches; neither P nor B_s is a universal localizer.

The remaining open target is to derive, from the carrier family $\{A_s : s \in P\}$, their punctures B_s, the external-neighbour structure, and the augmented forest quotient, either a hereditarily violating subset of $P \cup \bigcup_s A_s$ or a feasible exchange avoiding a forced-cycle edge. The witnesses show the violating set need not be P, B_s, or T_F; no single localization is available.

Computationally, Lemma A holds on every instance up to $n = 10$ (an exhaustive forced-core scan over $n = 7, \ldots, 10$ found no cyclic forced core; the forced core was always a forest).[1]

4.2 Lemma B: cycle-erasing descent

Lemma B asks that a CES toggle exist for every selected cycle that has an avoidable incidence (Lemma A guarantees every selected cycle is of this kind). The natural metric refinement is formulated through residual augmenting paths. For such a cycle C let $d_0(C)$ be the minimum length of a residual augmenting path over the avoidable edges of C, and $d_{\mathrm{CES}}(C)$ the minimum length over CES residual paths.

Open Problem 4.8 (Lemma B)**.** Prove that every selected cycle containing an avoidable incidence admits a CES toggle; equivalently, $d_{\mathrm{CES}}(C) < \infty$ for every such cycle C.

A shortest-path-only formulation of Lemma B (that some globally shortest residual augmenting path is already CES) held exhaustively through $n = 10$ but was falsified at $n = 11$ by a sealed holdout (six 4-cycle witnesses at which a shortest bad path destroys C but creates one crossing cycle, the cycle-erasing reroute appearing only at length $d_0 + 2$). Two refinements survive those

[1] An $n = 11$ run over 994,607 instances reported zero cyclic cores and maximum forced-path length 8; that run is recorded in a preserved intermediate report, incompletely ledgered, and not independently reproducible from a completion artifact. We therefore do not claim exhaustive $n = 11$ verification of Lemma A.

exact witnesses and are supported *in sample only*: the metric bound $d_{\mathrm{CES}}(C) \le d_0(C) + 2$, and a terminal-slide mechanism producing the CES move by a two-edge extension of a shortest path; at $n = 11$ the gap distribution over the 8,202,633 selected cycles was $\{0 : 8{,}202{,}627,\ 2 : 6\}$. These are refinements of an open lemma, not theorems; a prospective $n = 12$ holdout was aborted without a verdict. We record the following conditional step, whose hypotheses the terminal-slide mechanism is meant to supply.

Proposition 4.9 (Terminal-slide criterion)**.** *Let $a = (s, h)$ be an avoidable incidence, Q_0 a simple residual augmenting path from the release ending at an H-terminal t, and $J_0 = \mathrm{toggle}(Q_0)$ feasible, with final added edge (u, t). Suppose (H1) C is not a cycle of J_0; (H2) J_0 has exactly one cycle Z not already present; (H3) (u, t) lies on Z; (H4) (u, h) is an unselected incidence with $u \neq s$ and $\deg_{J_0-(u,t)}(h) < c(h)$; (H5) u and h lie in different components of $J_0 - (u, t)$. Then $J^+ = J_0 - (u, t) + (u, h)$ is feasible, still omits a, and is CES for C.*

Proof. Feasibility is termwise: t loses one incidence; u swaps (u, t) for (u, h), keeping selected degree 2; h gains one incidence but $\deg_{J_0-(u,t)}(h) < c(h)$ by (H4); all other vertices are unchanged. Since $(u, h) \neq a$, the released a stays unselected. For cycles: by (H2)–(H3), Z is the only new cycle and $(u, t) \in Z$, so $J_0 - (u, t)$ has only cycles already present in the pre-toggle selection; adding (u, h) across two different components (H5) creates no cycle. Hence J^+ destroys C and adds no new cycle: it is CES. □

The open content of Lemma B is to prove that, for every positive-gap cycle, a shortest residual path exists whose terminal configuration satisfies (H1)–(H5)—an argument that must invoke residual shortest-path structure and hereditary tightness rather than any single empirical normal form.

5 Toward the general case

We turn to $o > 0$. Two monotonicities are immediate.

Lemma 5.1. *For any vertex set T: $\ell(G) \ge \ell(G - T)$ and $\tau(G - T) \ge \tau(G) - |T|$. In particular for a minimum odd-cycle transversal T with $B = G - T$ bipartite, $\ell(G) \ge \ell(B)$ and $\tau(B) \ge \tau(G) - o$.*

Proof. A spanning linear forest of $G - T$ extends to one of G by adjoining the vertices of T as trivial paths, so $\ell(G) \ge \ell(G - T)$. Deleting $|T|$ vertices lowers the minimum FVS by at most $|T|$, so $\tau(G - T) \ge \tau(G) - |T|$. □

Lemma 5.2 (Edge in the maximum bipartite subgraph)**.** *If G is connected and nontrivial, a maximum induced bipartite subgraph B has an edge; more generally $b(G) \ge \alpha(G) + 1$ where α is the independence number.*

Proof. If B were edgeless then $b(G) = \alpha(G)$. Take a maximum independent set I; since G is connected and nontrivial there is a vertex $v \notin I$ with a neighbour in I, and $I \cup \{v\}$ induces a star, which is bipartite, of size $|I| + 1 > b(G)$, contradicting maximality. □

Theorem 5.3 (Conditional coefficient-two bound)**.** *Assume the bipartite case of Conjecture 1.1 ($\ell(B) \ge 2\tau(B) + 1$ for every connected nontrivial bipartite B). Then for every connected nontrivial graph G,*

$$\ell(G) + 2\,o(G) \;\ge\; 2\,\tau(G) + 1.$$

Proof. Let T be a minimum odd-cycle transversal, $|T| = o$, and $B = G - T$ a maximum induced bipartite subgraph. By Lemma 5.1, $\ell(G) \geq \ell(B)$ and $\tau(B) \geq \tau(G) - o$. Applying the bipartite case componentwise to B (an edgeless component has $\ell = 0 = 2\tau$), $\ell(B) \geq 2\tau(B) + c_e$, where c_e is the number of components of B with an edge. By Lemma 5.2, $c_e \geq 1$. Hence

$$\ell(G) \geq \ell(B) \geq 2\tau(B) + c_e \geq 2(\tau(G) - o) + 1,$$

so $\ell(G) + 2o \geq 2\tau(G) + 1$. □

Theorem 5.3 has coefficient 2 on o; the target has coefficient 1. We isolate a single-vertex property that closes this gap.

Definition 5.4 (SV)**.** Say SV holds if for every connected nonbipartite finite simple graph G there is a vertex t lying in some minimum odd-cycle transversal of G with $p(G) \leq p(G - t)$, equivalently $\ell(G) \geq \ell(G - t) + 1$.

Note t lies in a minimum odd-cycle transversal iff $o(G - t) = o(G) - 1$.

Theorem 5.5 (Single-vertex reduction)**.** *If SV holds and the bipartite case of Conjecture 1.1 holds, then $\ell(G) + o(G) \geq 2\tau(G) + 1$ for every connected nontrivial G; i.e. Conjecture 1.1 holds in full.*

Proof. Strong induction on $|V(G)|$. If G is bipartite, $o(G) = 0$ and the bipartite case gives $\ell(G) \geq 2\tau(G) + 1 = 2\tau(G) + o(G) + 1$. Otherwise G is connected nonbipartite; take t as in SV, so $o(G - t) = o(G) - 1$ and $\ell(G) \geq \ell(G - t) + 1$. Also $\tau(G - t) \geq \tau(G) - 1$. The parameters ℓ, o, τ are additive over connected components, and to each edge-containing component K_i of $G - t$ apply either the induction hypothesis (if nonbipartite, $|K_i| < |G|$) or the bipartite case (if bipartite); both give $\ell(K_i) + o(K_i) \geq 2\tau(K_i) + 1$. Isolated components contribute 0. Summing over the $c_+(G - t)$ edge-containing components,

$$\ell(G - t) + o(G - t) \;\geq\; 2\tau(G - t) + c_+(G - t).$$

Finally $c_+(G - t) \geq 1$: G has an odd cycle C; if $t \notin C$ then C survives, and if $t \in C$ then $C - t$ is a path with an edge, so $G - t$ has an edge. Combining,

$$\ell(G)+o(G) \;\geq\; (\ell(G-t)+1)+(o(G-t)+1) \;\geq\; (2\tau(G-t)+c_+)+2 \;\geq\; 2(\tau(G)-1)+1+2 = 2\tau(G)+1.$$

□

SV is a sufficient route, not a necessary one: a failure of SV would not refute Conjecture 1.1. Computationally, SV holds for every connected nonbipartite graph on ≤ 10 vertices (Section 6).

6 Computational verification

Methodology. All computational claims here come from a preregistered protocol. A single exact checker `phase0.c` computes f (maximum induced forest, by a subset scan with union-find acyclicity test), b (maximum induced bipartite subgraph, by a subset scan with a 2-colouring test), and p (minimum path cover $= n -$ maximum spanning linear forest, by a Hamiltonicity/DP bitmask) exactly for $n \leq 11$. The checker source was frozen and SHA-256 hashed before any run; a gold suite (invariants on K_{10}, $K_{5,5}$, $K_{1,9}$, complete multipartite families, C_{10}, and others) and a 995-graph regression over $n \leq 7$ passed before launch. Graph enumeration used `nauty geng -c` [4]. Every low-slack graph was re-checked by a second, independently coded engine (`phase0_verify2.py`, a pure-Python bitmask implementation sharing no code with the C checker). Any bound violation would halt the run with the graph and witness sets; none occurred.

Direct verification of Conjecture 1.1.

- $2 \le n \le 10$: every connected graph was enumerated, with 0 violations at every order. The connected-graph counts matched the known values exactly: $n = 2$ through 7 give $1, 2, 6, 21, 112, 853$ (total 995, the regression layer); $n = 8$ gives 11,117; $n = 9$ gives 261,080; $n = 10$ gives 11,716,571. The layers $n = 8, 9$ were run under the same re-frozen checker as $n = 11$ (re-golded before use), so the whole range sits under one dual-engine protocol. At $n = 10$ the minimum slack is 0, with 115,154 low-slack graphs (slack ≤ 1) of which 4,960 meet the bound with equality ($2f = p + b + 1$); all low-slack graphs of every order $n \le 10$, plus a 10,005-graph sample at $n = 10$, were re-verified by the second engine with 0 violations. (Checker `phase0.c`, freeze hashes `9f5cc322` for the original $n \le 10$ runs and `bc7f614d` for the re-frozen $n = 8, 9$ and $n = 11$ runs.)
- $n = 11$: all 1,006,700,565 connected graphs on 11 vertices were enumerated, sharded into 192 residue classes whose counts sum to exactly 1,006,700,565; there were 0 violations, minimum slack 0, and 3,853,461 low-slack graphs, of which 96,949 meet the bound with equality (and 3,756,512 have slack 1). All 3,853,461 low-slack graphs plus a 9,968-graph sample — 3,863,429 graphs in all — were re-verified by the second engine, with 0 violations. (Checker `phase0.c`, re-freeze hash `bc7f614d`; wall time 18.1 hours on an 8-core machine.)

Thus Conjecture 1.1 holds, with no exceptions, for every connected graph on at most 11 vertices.

Verification of SV. The single-vertex property was checked directly: for each connected nonbipartite G the eligible set $\{t : o(G - t) = o(G) - 1\}$ was computed and tested for a vertex with $p(G) \le p(G - t)$. SV held for all 272,209 connected nonbipartite graphs with $3 \le n \le 9$, and for all 11,712,539 connected nonbipartite graphs on $n = 10$; 0 violations in every case.

Status of the lemma-level runs. Lemma A (forced-core acyclicity) was verified exhaustively for $n \le 10$ (no cyclic forced core). Lemma B's shortest-only form was verified exhaustively through $n \le 10$ and *falsified* at $n = 11$ by a sealed holdout (six witnesses), after which the surviving metric and terminal-slide refinements were supported only in sample (Section 4); a prospective $n = 12$ holdout was aborted without a verdict.

7 Related work

The nearest neighbour is the work of Lavrov and Vandenbussche [1]. They do not characterize path covers; they *conjecture* a Hall-type sufficient condition under which one side X of a bipartite graph can be covered by at most $\Lambda\text{-def}(G)$ vertex-disjoint paths, and they *prove* the conjecture for forests, for graphs of maximum degree 3, and for regular graphs of high girth. Their Λ-deficiency and the demand-2 deficiency δ_2 of our incidence model (Section 3.1) are related. Taking the model with *all capacities* 2 (no capacity-1 pair), write, for a vertex set S, $\mathrm{supp}(S)$ for the number of H-vertices with at least one neighbour in S, and $\Lambda(S)$ for the number with at least two; then the two deficiencies satisfy

$$\delta_2(S) \;+\; \Lambda\text{-def}(S) \;=\; \mathrm{supp}(S) \;-\; |S|.$$

[2] The obstacle to transferring their machinery directly is that Λ-def is neither submodular nor supermodular, so the tight-set lattice (Lemma 3.6) that drives our carrier analysis has no counterpart

[2] This identity is stated here as the bridge between the two deficiency measures; it is not derived from the proof documents of this project and we do not claim it as one of the results verified here.

on their side. Separately, the parameters $f = n - \tau$ and the decycling number τ sit in the large induced-forest / feedback-vertex-set literature, and $b = n - o$ with o in the odd-cycle-transversal literature; the novelty of the deficiency form (Theorem 2.1) is that it couples all three of τ, o, and the linear-forest parameter ℓ in a single inequality.

8 Data availability

The frozen checker `phase0.c`, the independent verifier `phase0_verify2.py`, the per-shard enumeration counts, the second-engine re-verification logs, and the compressed low-slack witness sets are archived with the manuscript. *[Repository URL and Zenodo DOI to be inserted.]*

Acknowledgments

The mathematics and the computations in this paper were developed with AI assistance (Claude models); all results were independently cross-checked computationally.

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